

Popular Vote

Requirements Document

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Context

CTX1: Description

Popular Vote is a web-based application that allows participants in a shared session to privately submit questions about a topic. The system collects these questions, uses AI to cluster similar ones, and condenses them into a clear, organized set of representative queries. Instead of a moderator fielding dozens of scattered, redundant, or never-verbalized questions, session hosts receive a structured view of what the group actually wants to understand. The goal is to surface collective concerns without putting pressure on individuals to speak up.

CTX2: Scope

- **CTX2.1:** Popular Vote is scoped to structured, host-moderated sessions where participants submit questions and a host views the clustered results.
- **CTX2.2:** The following are explicitly out of scope for the current version:
 - **CTX2.2.1:** Participant-facing display of individual submissions from other participants
 - **CTX2.2.2:** Real-time or live-updating clusters as submissions come in
 - **CTX2.2.3:** User accounts, authentication, or persistent profiles
 - **CTX2.2.4:** Moderation tools for filtering individual submissions before clustering
 - **CTX2.2.5:** Analytics, session history, or export features
 - **CTX2.2.6:** Multi-host or co-moderator sessions

CTX3: User Classes

- **CTX3.1:** Session Host. The individual who creates and manages a session. The Host is responsible for opening the session, sharing the session code with participants, and initiating the AI clustering step. The Host is the only user who views the condensed, organized output of clustered questions. Each session has exactly one Host.
- **CTX3.2:** Participant. Any individual who joins an active session using a session code and privately submits one or more questions. Participants cannot see other participants' individual submissions, but can view the clustered output once the Host releases it. Participation requires no account or identity disclosure.

CTX4: Operating Environment

- **CTX4.1:** The application runs in a standard web browser on any desktop or mobile device. No installation is required.
- **CTX4.2:** Both Hosts and Participants must have an active internet connection for the duration of the session.
- **CTX4.3:** AI clustering is performed by a third-party language model API accessed over the internet. Clustering is not available offline.

CTX5: Assumptions and Dependencies

- **CTX5.1:** The Host is responsible for communicating the session code to participants through an external channel (e.g., verbal announcement, projected display, or message). The system does not handle participant outreach.
- **CTX5.2:** Participants are assumed to be acting in good faith. The system does not enforce content moderation in the current version.
- **CTX5.3:** The third-party AI API used for clustering is assumed to be available and responsive during a session's clustering step.

Functional Requirements

FR1: Session Creation (Host)

- **FR1.1:** The system shall allow a user to create a new session without requiring an account.
- **FR1.2:** Upon creation, the system shall generate a short, unique session code that participants can use to join.
- **FR1.3:** The system shall display the session code prominently to the Host so it can be shared with participants.
- **FR1.4:** The system shall present the Host with a live count of participants who have joined the session.
- **FR1.5:** The system shall allow the Host to close the submission window at any time, preventing further question submissions.
- **FR1.6:** The system shall allow the Host to trigger AI clustering only after the submission window has been closed.
- **FR1.7:** The system shall allow the Host to add tags to the session to give the AI some context about the session.

FR2: Session Joining (Participant)

- **FR2.1:** The system shall allow a user to join an existing session by entering a session code. No account is required.
- **FR2.2:** The system shall reject join attempts using invalid or expired session codes and display a clear error message.

- **FR2.3:** The system shall reject join attempts for sessions whose submission window has been closed and inform the user that the session is no longer accepting submissions.

FR3: Question Submission (Participant)

- **FR3.1:** The system shall allow a participant to submit a text-based question after joining a session.
- **FR3.2:** The system shall allow a participant to submit more than one question per session.
- **FR3.3:** The system shall not display any participant's submitted questions to other participants.
- **FR3.4:** The system shall not attach any identifying information to a submitted question.
- **FR3.5:** The system shall reject empty submissions and inform the participant that a question is required.
- **FR3.6:** The system shall enforce a maximum character limit per question and notify the participant when the limit is reached.
- **FR3.7:** The system shall confirm to the participant that their question was received after each successful submission.
- **FR3.8:** The system shall allow a participant to delete their submission in case they submit something on accident or unrelated to the topic.

FR4: AI Clustering

- **FR4.1:** When the Host initiates clustering, the system shall submit all collected questions to an AI model and request a condensed set of representative queries.
- **FR4.2:** The system shall display a processing indicator to the Host while clustering is underway.
- **FR4.3:** The system shall present the clustered output to both the Host and all active Participants in an organized format, grouping similar questions and displaying a condensed representative query for each cluster.
- **FR4.4:** The system shall indicate to the Host approximately how many submissions contributed to each cluster.
- **FR4.5:** If the AI clustering step fails, the system shall notify the Host and allow them to retry without losing the submitted questions.
- **FR4.6:** If no questions were submitted before the Host initiates clustering, the system shall inform the Host and not make an API call.
- **FR4.7:** The system should allow the Host to delete a cluster.
- **FR4.8:** If a Participants submission contains multiple questions, the AI shall treat them as separate questions to cluster.

FR5: Session Lifecycle

- **FR5.1:** The system shall allow the Host to end a session after clustering is complete.

- **FR5.2:** When a session ends, the system shall prevent further participant access to that session code.
- **FR5.3:** If a participant attempts to submit a question after the submission window has been closed, the system shall notify them that the window is no longer open.
- **FR5.4:** At the end of a session, the system shall display a condensed summary of the clustering results to the Host and all active Participants, including the representative query for each cluster and the approximate number of submissions per cluster.

Non-Functional Requirements

- **NFR1:** Submission Anonymity. The system shall not store, log, or transmit any personally identifiable information associated with a submitted question. No IP address, device identifier, or session cookie shall be linked to individual question records.
- **NFR2:** Session Privacy. A participant may only access a session via its unique session code. Sessions shall not be discoverable or listable by any user.
- **NFR3:** Submission Responsiveness. Question submissions shall be acknowledged to the participant within two seconds of the triggering action under normal network conditions.
- **NFR4:** Clustering Latency. The system shall initiate the clustering API call within one second of the Host triggering the clustering step. The total clustering round-trip time is subject to the third-party API's response time, which is outside the system's control.
- **NFR5:** Data Retention. Submitted questions and clustering results shall not be persisted after the session ends. No session data shall be retained on the server once the Host closes the session.
- **NFR6:** Concurrent Submissions. The system shall correctly accept and store submissions from multiple participants submitting simultaneously within the same session.
- **NFR7:** The system should have rate limiting to prevent Participants from spamming submissions.

Note on omitted categories: Maintainability logging is omitted because the absence of persistent data and accounts makes post-hoc debugging a low-priority concern for the current version. Reliability through redundancy (e.g., failover infrastructure) is also omitted, as the session-based, stateless nature of the system limits the consequences of a transient outage.

Data Requirements

DR1: Active Session State

- **DR1.1:** The system must maintain a unique session code and its associated active or closed status.
- **DR1.2:** The system must maintain the count of participants who have joined the session.
- **DR1.3:** The system must track whether the submission window is open or closed.

- **DR1.4:** The system must track whether clustering has been initiated and whether results have been returned.

DR2: Question Submissions

- **DR2.1:** The system must store the text content of each submitted question for the duration of the session.
- **DR2.2:** The system must associate each submission with its session without associating it with any individual participant.

DR3: Clustering Output

- **DR3.1:** The system must store the AI-generated clustered output for the duration of the session so the Host can review it.
- **DR3.2:** The system must track the approximate count of submissions that contributed to each cluster.

Constraints

- **CON1:** AI clustering depends on an external language model API. The quality, latency, and availability of clustering results are subject to that API's terms and capabilities.
- **CON2:** The system does not support offline use. Both session creation and question submission require an active internet connection.
- **CON3:** The system does not authenticate Hosts or Participants. There is no mechanism to verify that a Host is who they claim to be, or to prevent a single individual from submitting multiple questions.
- **CON4:** Content moderation is not supported in the current version. The system transmits all submitted questions to the AI API as-is.
- **CON5:** Session data is ephemeral. Once a session ends, all submitted questions and clustering results are discarded and cannot be recovered.

Future Directions

- **FD1:** Host-Controlled Release Timing. Currently the clustered output becomes visible to Participants immediately when clustering completes. A future version could give the Host the ability to review and optionally edit or suppress specific clusters before releasing the output to Participants, adding a moderation layer between clustering and display.
- **FD2:** Live Clustering. Instead of triggering clustering once at the end, the system could cluster incrementally as questions arrive, giving the Host a continuously updated view of emerging themes during an active session.
- **FD3:** Content Moderation Layer. Before clustering, a moderation step could flag or filter submissions that contain inappropriate content, making the system safer for use in public-facing or school environments.

- **FD4:** Session History and Export. Hosts could optionally save and export the clustered results after a session, enabling follow-up communication or documentation. This would require an opt-in persistence model.
- **FD5:** Host Accounts and Session Management. Persistent host accounts would allow session history, reusable session templates, and the ability to reconnect to a session that was interrupted.
- **FD6:** Question Upvoting. Participants could be given the ability to upvote questions submitted by others (while remaining anonymous), providing an additional signal of collective priority beyond clustering alone.
- **FD7:** QR Code Generation. The system should also generate a QR code for the session that the Host can share to make Participants on mobile join the session faster.

Appendix A: Glossary

- **A1:** Session. A bounded event created by a Host, during which Participants may submit questions and the Host may trigger AI clustering.
- **A2:** Session Code. A short, unique alphanumeric code generated at session creation that Participants use to join the session.
- **A3:** Submission Window. The period during a session when Participants are permitted to submit questions. The Host controls when the window opens and closes.
- **A4:** Clustering. The AI-driven process of grouping semantically similar questions and generating a condensed, representative query for each group.
- **A5:** Cluster. A group of submitted questions that the AI has determined share a common underlying concern or theme, represented by a single condensed query in the output.